

# Solutions to "Algebraic Geometry and Arithmetic Curves"

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## 1 Some topics in commutative algebra

### 1.1 Tensor products

**Exercise 1.1.1.** Let  $\{M_i\}_{i \in I}, \{N_j\}_{j \in J}$  be two families of modules over a ring  $A$ . Then

$$(\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j).$$

*Proof.* There is a chain of natural isomorphisms:

$$\begin{aligned} & (\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \\ & \cong \oplus_{i \in I} (M_i \otimes_A (\oplus_{j \in J} N_j)) \\ & \cong \oplus_{j \in J} \oplus_{i \in I} (M_i \otimes_A N_j) \\ & \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j). \end{aligned}$$

□

**Exercise 1.1.2.** There is a unique  $A$ -linear map

$$f: \text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that  $f(u \otimes v) = u \otimes v$ .

*Proof.* Consider the map  $g: \text{Hom}_A(M, M') \times \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$  sending  $(u, v)$  to  $u \otimes v$ . It is clearly bilinear, so factors uniquely through  $\text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N')$ .  $\square$

**Exercise 1.1.3.** Let  $M, N$  be  $A$ -modules and let  $i: M' \rightarrow M, j: N' \rightarrow N$  be submodules of  $M$  and  $N$ , respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N).$$

In particular,  $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/m\mathbb{Z}) = \mathbb{Z}/l\mathbb{Z}$ , where  $l = \gcd(m, n)$ .

*Proof.* Observe that

$$(M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N) \cong ((M \otimes_A N) / (\text{Im } i_N)) / (\text{Im } j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} & ((M \otimes_A N) / (\text{Im } i_N)) / (\text{Im } j_{M/M'}) \\ & \cong ((M/M') \otimes_A N) / (\text{Im } j_{M/M'}) \\ & \cong (M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that  $(m) + (n) = (l)$ .  $\square$

**Exercise 1.1.4.** Let  $M, N$  be  $A$ -modules and let  $B, C$  be  $A$ -algebras.

- (a) If  $M$  and  $N$  are finitely generated over  $A$ , then so is  $M \otimes_A N$ .
- (b) If  $B$  and  $C$  are finitely generated over  $A$ , then so is  $B \otimes_A C$ .
- (c) The converses of (a) and (b) are in general false.

*Proof.* Parts (a) and (b) are obvious. For part (c), just notice that  $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$  is the zero ring, but  $\mathbb{Q}$  is not finitely generated over  $\mathbb{Z}$ .  $\square$

**Exercise 1.1.5.** Let  $\rho: A \rightarrow B$  be a ring homomorphism,  $M$  an  $A$ -module, and  $N$  a  $B$ -module. There is a canonical isomorphism of  $A$ -modules

$$\text{Hom}_A(M, \rho_* N) \cong \text{Hom}_B(\rho^* M, N).$$

*Proof.* Let  $f: \text{Hom}_A(M, \rho_* N) \rightarrow \text{Hom}_B(\rho^* M, N)$  be the map sending  $u$  to the map  $f(u)$  s.t.  $f(u)(m \otimes b) = bu(m)$ . This is well-defined because if  $n \otimes c = m \otimes b$ , we may assume that there is some  $a \in A$  s.t.  $an = m$  and  $\rho(a)b = c$  and so  $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$ .

In addition, define  $g: \text{Hom}_B(\rho^* M, N) \rightarrow \text{Hom}_A(M, \rho_* N)$  to be the map s.t.  $g(u)(m) = u(m \otimes 1)$ . This is immediately seen to be well-defined, and we contest that this is an inverse for  $f$ . Indeed,  $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$  and  $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$ .  $\square$

**Exercise 1.1.6.** Let  $(N_i)_{i \in I}$  be a direct system of modules. Then for any  $A$ -module  $M$ , there exists a canonical isomorphism  $\varinjlim (N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$ .

*Proof.* Tensor-hom. □

Just kidding, it follows from the fact that  $\varinjlim (N_i \times M)$  is canonically isomorphic to  $(\varinjlim N_i) \times M$ , and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

**Exercise 1.1.7.** Let  $B$  be an  $A$ -algebra, and let  $M, N$  be  $B$ -modules. Then there exists a canonical surjection  $M \otimes_A N \rightarrow M \otimes_B N$ .

*Proof.* Do I really need to spell it out? □

## 1.2 Flatness

**Exercise 1.2.1.** Let  $M$  be an  $A$ -module and let  $I \subseteq \text{Ann}(m)$  be an ideal.

- (a) There is a natural  $A/I$ -module structure on  $M$ , and  $M \cong M \otimes_A A/I$ .
- (b) Let  $N$  be another  $A$ -module such that  $I \subseteq \text{Ann}(N)$ . The canonical homomorphism  $M \otimes_A N \rightarrow M \otimes_{A/I} N$  is an isomorphism.

*Proof.* For (a), observe that if  $x \in I$ , then  $am = (a + x)m$  for any  $a \in A$  and  $m \in M$ . This implies that the multiplication of elements of  $M$  by cosets  $a + I$  is well defined, hence we have an  $A/I$ -module structure on  $M$ . To see that this is the extension of scalars, recall that  $M \otimes_A A/I \cong M/(IM)$ , which is just  $M$ .

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} M \otimes_A N & \\ \cong (M \otimes_A A/I) \otimes_{A/I} N & \\ \cong M \otimes_{A/I} N, & \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a). □

**Exercise 1.2.2.** Let  $\rho: A \rightarrow B$  be a ring homomorphism,  $S$  a multiplicative subset of  $A$ , and  $T = \rho(S)$ . Then  $T$  is a multiplicative subset of  $B$ , and  $T^{-1}B \cong B \otimes_A S^{-1}A$ .

*Proof.* The first part is obvious. To see the last statement, let  $\iota: B \rightarrow T^{-1}B$  be the canonical embedding. Since  $\iota$  sends elements of  $T = \rho(S)$  to units, there is a canonical homomorphism  $\phi: S^{-1}A \rightarrow T^{-1}B$ . Since  $\otimes_A$  is a pushout in the category of  $A$ -algebras, there is a unique homomorphism  $B \otimes_A S^{-1}A \rightarrow T^{-1}B$  such that this diagram commutes:

$$\begin{array}{ccc} S^{-1}A \otimes B & \longleftarrow & S^{-1}A \\ \uparrow & \searrow & \downarrow \\ B & \longrightarrow & T^{-1}B. \end{array}$$

This is an isomorphism because of universal property of localizations.  $\square$

**Exercise 1.2.3.** Nakayama's lemma does not hold for modules that are not finitely generated.

*Proof.* Consider  $\mathbf{Q}$  as a module over  $\mathbf{Z}_{(p)}$ .  $\square$

**Exercise 1.2.4.** Let  $I$  be an ideal of a ring  $A$ . Then the following are equivalent:

- (a)  $A/I$  is flat over  $A$ .
- (b)  $I^2 = I$
- (c) There is an element  $e \in I$  such that  $e^2 = e$  and  $eA = I$ .

*Proof.* ((i) $\Rightarrow$ (ii)) Consider the exact sequence  $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ . Since  $A/I$  is flat, we have another exact sequence  $0 \rightarrow A/I \rightarrow A/I^2 \rightarrow 0$ , so  $I = I^2$ .

((ii) $\Rightarrow$ (iii)) I don't know how to do this morally. The easy way out is to use the form of Nakayama's Lemma proven by the determinant trick.

((iii) $\Rightarrow$ (i)) Observe that in any maximal ideal,  $I$  either localizes to  $(0)$  or  $(1)$ . In any case, it's flat at every maximal ideal, hence flat overall.  $\square$

## 1.3 Formal completion

**Exercise 1.3.1.** The usual topology on  $\mathbf{R}$  is not given by a subgroup filtration.

*Proof.* Use noggin for  $\geq 2$  seconds.  $\square$

# 2 General properties of schemes

## 2.1 Spectrum of a ring

**Exercise 2.1.1.** For  $k$  a field,  $\text{Spec}(k[[T]])$  is the space  $\{\xi, 0\}$  with generic point  $\xi$  and closed point  $0$ .

*Proof.* It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have  $f = \sum a_n T^n$  with  $a_N = 1$  for some  $N$  and  $a_n = 0$  whenever  $n \leq N$ , then we can pull out a common factor of  $T^N$ . We see that  $f$  is then a unit multiple of  $T^N$  and  $(f) = (T^N)$ . Clearly, this is only prime when  $N = 1$ . It is easy to see that  $k[[T]]$  is an entire ring, so the proposition is immediate.  $\square$

**Exercise 2.1.2.** Let  $\varphi: A \rightarrow B$  be a homomorphism of finitely generated algebras over a field  $k$ . The image of a closed point under  $\text{Spec } \varphi$  is a closed point.

*Proof.* If  $\text{Spec } \varphi$  takes the maximal ideal  $\mathfrak{m}$  into the ideal  $\mathfrak{p}$ , we get a diagram

$$k \rightarrow A/\mathfrak{p} \xrightarrow{\tilde{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12,  $B/\mathfrak{m}$  is a finite algebraic extension of  $k$ , so  $B/\mathfrak{m}$  (resp.  $A/\mathfrak{p}$ ) is finite over  $A/\mathfrak{p}$  (resp.  $k$ ). It then suffices to prove that  $A/\mathfrak{p}$  is a field. Any  $a \in A$  has an associated irreducible polynomial  $P = \sum a_i X^i \in k[x]$  such that  $P(a) = 0$ . This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of  $a$ .  $\square$

**Exercise 2.1.3.** Let  $k = \mathbf{R}$  be the field of real numbers and let  $A = k[X, Y]/(X^2 + Y^2 + 1)$ . Let  $x, y$  be the respective images of  $X, Y$  in  $A$ .

- (a) Let  $\mathfrak{m}$  be a maximal ideal of  $A$ . Then there are  $a, b, c, d \in k$  such that  $x^2 + ax + b, y^2 + cy + d \in \mathfrak{m}$ . Furthermore,  $\mathfrak{m}$  is generated by a single element  $f = \alpha x + \beta y + \gamma$  with at least one of  $\alpha, \beta$  nonzero.
- (b) The map  $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$  establishes a bijection between the subset  $\mathbf{P}(k^3) \setminus \{(0, 0, 1)\}$  of the projective space  $\mathbf{P}(k^3)$  and the set of maximal ideals of  $A$ .
- (c) Let  $\mathfrak{p}$  be a non-maximal prime ideal of  $A$ . Then the canonical homomorphism  $\varphi: k[X] \rightarrow A$  is finite and injective, and  $k[X] \cap \mathfrak{p} = 0$ . Let  $g \in \mathfrak{p}$  and let  $g^n + a_{n-1}g^{n-1} + \dots + a_0 = 0$  be an integral equation with  $a_i \in k[X]$ . Then  $a_0 = 0$ , and so  $\mathfrak{p} = 0$ .

*Proof.* First, notice that  $A/\mathfrak{m}$  is isomorphic to  $\mathbf{C}$ , since  $X^2 + Y^2 + 1$  does not split in  $k$ . Then the existence of  $a, b, c, d$  follows from the fact that any complex number satisfies a monic quadratic equation. Since  $\mathbf{C}$  is degree 2 over  $k$ , there are  $\alpha, \beta, \gamma$ , not all zero, such that

$$\alpha x' + \beta y' + \gamma = 0,$$

where  $x'$  and  $y'$  are the respective images of  $x$  and  $y$ . Since not all of them are zero, we may assume that  $\alpha$  is nonzero. Then we get an equation for  $x'$  in terms of  $y'$ , which we can then substitute into the equation  $x'^2 + y'^2 + 1 = 0$ . This new equation cannot split in  $k$  because that would imply a real root of  $x'^2 + y'^2 + 1 = 0$ , so we have that  $\mathfrak{m} = fA$ , where  $f$  is the ideal corresponding to the linear equation in  $x'$  and  $y'$ .

Part (b) holds, since a solution to  $X^2 + Y^2 + 1 = 0$  is determined (up to conjugacy) by an arbitrary complex number, which itself determines a linear equation with real coefficients.

That  $\varphi$  is finite and injective is clear, so consider the induced homomorphism  $\tilde{\varphi}: k[X]/\varphi^{-1}(\mathfrak{p}) \rightarrow A/\mathfrak{p}$ . Notice that  $B = k[X]/\varphi^{-1}(\mathfrak{p})$  is not a field, since  $\tilde{\varphi}$  is

still finite and injective, which implies that  $A/\mathfrak{p}$  is a field as well. Then  $B = k[X]$  and  $\mathfrak{p} \cap k[X] = 0$ . Finally, in the integral equation for  $g$ , notice that  $a_0$  must be a multiple of  $g$ , hence 0. Then the equation has a common factor of  $g$  which we may cancel, which gives a new equation with constant term  $a_1$ . This repeats indefinitely and so  $g = 0$ .  $\square$

**Exercise 2.1.4.** Let  $A$  be a ring.

- (a) Let  $\mathfrak{p}$  be a minimal prime ideal of  $A$ . Then every element of  $\mathfrak{p}A_{\mathfrak{p}}$  is nilpotent. In particular, every element of  $\mathfrak{p}$  is a zero divisor.
- (b) If  $A$  is reduced, then any zero divisor in  $A$  is an element of a minimal prime ideal. This is not true in general if  $A$  is not reduced.

*Proof.* Part (a) is obvious.

For (b), let  $x$  be any zero divisor and let  $\mathfrak{p}$  be a prime ideal minimal among those that contain  $x$ . If  $\mathfrak{p}$  is minimal, we are done. Otherwise, suppose that  $\mathfrak{q} \subset \mathfrak{p}$  is a minimal prime ideal not containing  $x$ . Since  $A$  is reduced, we must have  $\sqrt{(x)} \cap \text{Ann}(x) = \{0\}$ , so  $\mathfrak{q}$  must contain a nonzero element  $x' \in \text{Ann}(x)$ . Localizing by  $\mathfrak{q}$ , we see that  $x$  is sent to 0 and  $x'$  is sent to a nilpotent. This implies that there is some positive integer  $n$  and some element  $s \in A \setminus \mathfrak{q}$  such that  $s(x'^n - x) = 0$ , and it is easy to see that this means that  $sx^2 = 0 \in \mathfrak{q}$ , so  $x^2 \in \mathfrak{q}$ , contradiction.  $\square$

The fact that this is false for  $A$  not reduced comes is immediate from the fact that there can be non-nilpotent zero divisors in a ring.

**Exercise 2.1.5.** Let  $k$  be a field. Let  $\mathfrak{m}$  be a maximal ideal of  $k[T_1, \dots, T_n]$ .

- (a) Let  $P_1(T_1)$  be a generator of  $\mathfrak{m} \cap k[T_1]$  and  $k_1 = k[T_1]/(P_1)$ . Then we have an exact sequence

$$0 \rightarrow P_1 k[T_1, \dots, T_n] \rightarrow k[T_1, \dots, T_n] \rightarrow k_1[T_2, \dots, T_n] \rightarrow 0.$$

- (b) There exist  $n$  polynomials  $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$  such that  $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$  for all  $i \leq n$ . In particular,  $\mathfrak{m}$  is generated by  $n$  elements.

*Proof.* Part (a) is obvious. In part (b), we proceed by induction. The statement is obvious in the single-variable case, so suppose that it holds for  $n - 1$ , with  $n$  an integer at least 2. Let  $k_1$  be as in (a) and observe that it is a field, so the hypotheses and the exact sequence from (a) give a generating set  $(P_2, \dots, P_n)$  for the image of  $\mathfrak{m}$  in  $k_1[T_2, \dots, T_n]$ . This pulls back to a set of generators  $(P_1, \dots, P_n)$  for  $\mathfrak{m}$ , and it is easy to see that the desired properties hold.  $\square$

**Exercise 2.1.6.** Any closed subspace of a quasicompact space is itself quasicompact. If  $A$  is a ring,  $\text{Spec}(A)$  is quasicompact.

*Proof.* Let  $X$  be a topological space and let  $Y \subseteq X$  be any closed subset. If  $\{U_i\}$  is an open covering of  $Y$ ,  $\{U_i \cup \complement_X Y\}$  is an open covering of  $X$ . Taking a finite subcovering of this extended covering yields a finite subcovering of the original covering.

For the second statement, suppose that  $\{U_i\}$  is an open covering of  $X = \operatorname{Spec} A$ . For each  $i$ , let  $J_i$  be the ideal such that  $\complement_X U_i = V(J_i)$ . We see that  $\emptyset = \bigcap V(J_i) = V(\sum J_i)$ , so there is a finite family of  $f_k \in J_k$  such that  $\sum f_k = 1$ . Then  $\sum J_k = 1$  and  $\{U_k\}$  is a finite subcovering.  $\square$

The next exercise is a bit open ended in character, so the proof label is changed to reflect that.

**Exercise 2.1.7.** Describe the closed subsets of  $\operatorname{Spec}(\mathbf{Z}[T])$ . Show that they are not all principal.

*Answer.* The closed subsets of  $\operatorname{Spec}(\mathbf{Z}[T])$  can be broken up into three families: points, lines, and the generic point. The points are simply maximal ideals; in  $\mathbf{Z}[T]$  these are ideals of the form  $(p, P)$ , where  $p$  is a non-zero prime number and  $P$  is an irreducible polynomial. The lines are just principal ideals; I call them lines because they determine a family of points that looks kind of like a line. Finally, there is the generic point  $(0)$ . The fact that the closed subsets are not all principal follows from the fact that the maximal ideals have 2 generators.  $\square$

**Exercise 2.1.8.** Let  $\varphi: A \rightarrow B$  be an integral homomorphism.

- (a) The map  $\operatorname{Spec} \varphi: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$  sends closed points to closed points, and any point in the preimage of a closed point is also a closed point.
- (b) Let  $\mathfrak{p} \in \operatorname{Spec} A$ . The canonical homomorphism  $A_{\mathfrak{p}} \rightarrow B \otimes_A A_{\mathfrak{p}}$  is integral.
- (c) Let  $T = \varphi(A \setminus \mathfrak{p})$  and suppose that  $\varphi$  is injective. Then  $T$  is a multiplicative subset of  $B$  and  $B \otimes_A A_{\mathfrak{p}} = T^{-1}B \neq 0$ . In particular,  $\operatorname{Spec} \varphi$  is surjective if  $\varphi$  is integral and injective.

*Proof.* For part (a), suppose that  $\mathfrak{m}$  is a closed point of  $\operatorname{Spec} B$  and let  $\varphi^{-1}(\mathfrak{m}) = \mathfrak{m}'$ . Then we have an integral homomorphism  $A/\mathfrak{m}' \hookrightarrow B/\mathfrak{m}$ . Given  $a \in A/\mathfrak{m}'$ , the inverse  $b \in B/\mathfrak{m}$  of its image under  $\varphi$  satisfies an integral equation  $a_0 + a_1 b + \dots + b^n = 0$ . Multiplying by an appropriate power of  $a$ , we have  $a_0 a^n + \dots + 1 = 0$ , so  $a$  is a unit. On the other hand, let  $\mathfrak{m}$  be a closed point of  $\operatorname{Spec} A$  and suppose that  $\varphi(\mathfrak{m}) \subseteq \mathfrak{m}'$  for some  $\mathfrak{m}' \in \operatorname{Spec} B$ . Again, we have an integral homomorphism  $A/\mathfrak{m} \hookrightarrow B/\mathfrak{m}'$ . Since  $A/\mathfrak{m}$  is a field, this is an algebraic extension.

Parts (b) and (c) are easy, but somewhat annoying to write out.  $\square$

**Exercise 2.1.9.** Let  $k$  be a field and  $A$  be a finitely generated  $k$ -algebra.

- (a) Suppose in addition that  $A$  is a finite over  $k$ . Then  $\operatorname{Spec} A$  is a finite set, with cardinality bounded above by  $\dim_k A$ . Furthermore, every prime ideal of  $A$  is maximal.

(b) The cardinality of  $\text{Spec}(k[T_1, \dots, T_d])$  is infinite when  $d \geq 1$ .

(c) The set  $\text{Spec } A$  is finite iff  $A$  is finite over  $k$ .

*Proof.* For part (a), the second statement is obvious. To prove the first one, let  $\{a_1, \dots, a_n\}$  be a basis for  $A$  over  $k$ . We see that there is a family of maximal ideals  $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ , where  $\mathfrak{m}_i$  is the largest ideal not containing  $a_i$ . If  $\mathfrak{m}$  is any other maximal ideal, there is some  $i$  such that  $a_i \notin \mathfrak{m}$ , so  $\mathfrak{m} = \mathfrak{m}_i$ .

In (b), it suffices to prove it when  $d = 1$ . In this case, it's just Euclid's proof of the infinitude of primes.

Part (c) follows from parts (a) and (b) after an easy application of Noether normalization.  $\square$

## 2.2 Ringed topological spaces

Let  $X$  be a topological space.

**Exercise 2.2.1.** Let  $A$  be a non-trivial abelian group and let  $\mathcal{A}_X$  be the constant presheaf of  $A$  on  $X$ . The associated sheaf  $\mathcal{A}_X^\sharp$  is the sheaf of locally constant functions valued in  $A$ . In particular,  $\mathcal{A}_X$  is a sheaf if and only if every non-empty open subset of  $X$  is connected.

*Proof.* Let  $\mathcal{F}$  be the sheaf of locally constant  $A$ -valued functions on  $X$  and let  $\varphi: \mathcal{A}_X \rightarrow \mathcal{F}$  be the evident morphism. It is easy to see that the stalks of  $\mathcal{F}$  are all  $A$ . Let  $\mathcal{G}$  be any sheaf and let  $\alpha: \mathcal{A}_X \rightarrow \mathcal{G}$  be any morphism. If  $f$  is a locally constant function on  $U$ , we may split  $U$  into disjoint open sets  $U_a = f^{-1}(a)$  for  $a \in f(U)$ . We define  $\tilde{\alpha}(U): \mathcal{F} \rightarrow \mathcal{G}$  to be the map sending  $f$  to the regluing of the sections  $\alpha(U)(a)|_{U_a}$ . It is easy to see that this defines a morphism of sheaves and that  $\tilde{\alpha} \circ \varphi = \alpha$ . Since  $\varphi$  is surjective,  $\tilde{\alpha}$  is the unique such morphism.  $\square$

**Exercise 2.2.2.** Let  $\mathcal{F}$  be a sheaf on  $X$ . Let  $s, t \in \mathcal{F}(X)$ . Then the set of  $x \in X$  such that  $s_x = t_x$  is open in  $X$ .

*Proof.* If  $s_x = t_x$  at some point  $x \in X$ , there must be a neighborhood  $U_x$  of  $x$  such that  $s_y = t_y$  for all  $y \in U_x$ . Taking the union of all such  $U_x$  gives the result.  $\square$

**Exercise 2.2.3.** Let  $X$  be a fixed topological space. Let  $\mathcal{F}$  be a presheaf on  $X$ . We keep the notation introduced just before Lemma 2.7.

- (a) Let  $\mathcal{U} = \{U_i\}$  be an open covering of  $X$ . Let  $\mathcal{F}_{\mathcal{U}}(X) = \text{Ker } d_1$ . For any open subset  $W$  of  $X$ , we define in the same way a group  $\mathcal{F}_{\mathcal{U}}(W)$  relative to the covering  $\{W \cap U_i\}$  of  $W$ . Then  $\mathcal{F}_{\mathcal{U}}$  is a presheaf on  $X$  and we have a morphism of presheaves  $\mathcal{F} \rightarrow \mathcal{F}_{\mathcal{U}}$ .
- (b) Let  $\mathcal{V} = \{V_j\}$  be another open covering of  $X$ . We define a partial ordering:  $\mathcal{V} \preceq \mathcal{U}$  if  $\mathcal{V}$  is finer than  $\mathcal{U}$ . Then there is a canonical homomorphism  $\mathcal{F}_{\mathcal{U}}(W) \rightarrow \mathcal{F}_{\mathcal{V}}(W)$  for every open subset  $W$  if  $\mathcal{V} \preceq \mathcal{U}$ .

- (c) The ordering  $\preceq$  makes the set of open coverings of  $X$  into a direct system.  
Let

$$\mathcal{F}'(W) = \varinjlim_{\mathcal{V}} \mathcal{F}_{\mathcal{V}}(W),$$

the direct limit being taken over the open coverings of  $W$ . Then  $(\mathcal{F}')'$  is the sheaf associated to  $\mathcal{F}$ .

*Proof.* Part (a) is clear. The homomorphisms in part (b) are given by the restrictions in the natural way.

Part (c) is quite nasty to write out so I won't do it.  $\square$

**Exercise 2.2.4.** Let  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$  be an exact sequence of sheaves on  $X$ . Then the sequence of abelian groups  $0 \rightarrow \mathcal{F}'(X) \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}''(X)$  is exact.

*Proof.* This follows from the exact sequence induced on stalks.  $\square$

**Exercise 2.2.5.** Fix a sheaf  $\mathcal{G}$  on  $X$  and a closed point  $x_0 \in X$ . Define a presheaf  $\mathcal{F}$  by  $\mathcal{F}(U) = \mathcal{G}(U)$  if  $x_0 \notin U$  and  $\mathcal{F}(U) = \{s \in \mathcal{G}(U) : s_{x_0} = 0\}$  otherwise. Then  $\mathcal{F}$  is a sheaf and  $\text{Supp } \mathcal{F} = \text{Supp } \mathcal{G} \setminus \{x_0\}$ .

*Proof.* I think I'm missing something because this seems really immediate  $\square$

**Exercise 2.2.6.** Let  $f: X \rightarrow Y, g: Y \rightarrow Z$  be continuous maps,  $\mathcal{F}$  a sheaf on  $X$  and  $\mathcal{H}$  a sheaf on  $Z$ . Then

$$f^{-1}(g^{-1}\mathcal{H}) = (g \circ f)^{-1}\mathcal{H}, g_*(f_*\mathcal{F}) = (g \circ f)_*\mathcal{F}.$$

*Proof.* The second part is obvious. For the first one, notice that the assignment

$$\mathcal{F} \mapsto \left( U \mapsto \varprojlim_{V \supseteq f(U)} \mathcal{F}(U) \right)$$

is functorial, so we have a morphism  $f^{-1}(g^{-1}\mathcal{H}) \rightarrow (g \circ f)^{-1}\mathcal{H}$ . It is easy to check that this restricts to isomorphisms on the stalks, so it is an isomorphism. We can say this is equality because the isomorphism is sufficiently special.  $\square$

**Exercise 2.2.7.** Let  $\mathcal{B}$  be a base of open subsets for a topological space  $X$ . Let  $\mathcal{F}, \mathcal{G}$  be two sheaves on  $X$ . Suppose that for every  $U \in \mathcal{B}$ , there is a homomorphism  $\alpha(U): \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  which is compatible with restrictions. Then there is a unique extension  $\alpha: \mathcal{F} \rightarrow \mathcal{G}$  to a homomorphism of sheaves. Moreover, if  $\alpha(U)$  is surjective (resp. injective) for every  $U \in \mathcal{B}$ , then  $\alpha$  is surjective (resp. injective).

*Proof.* Let  $U$  be any open set and  $s$  a section of  $\mathcal{F}$  over  $U$ . If  $\{U_i\}$  is a cover of  $U$  by elements of  $\mathcal{B}$ , we have a corresponding family  $\{s_i\}$  of sections of  $\mathcal{F}$  given by restrictions of  $s$ . Since  $\alpha$  is compatible with restrictions, we obtain a family  $\{\alpha(U_i)(s_i)\}$  that agrees on intersections, which reglues into a section  $\alpha(s)$  of  $\mathcal{G}$ . It is easy to see that this is a morphism of sheaves, and the uniqueness of this extension follows from the uniqueness of gluing. The statement about surjectivity and injectivity is obvious.  $\square$

**Exercise 2.2.8.** Let  $\{U_i\}$  be an open covering of  $X$ . Suppose we are given a family of sheaves  $\mathcal{F}_i$  on  $U_i$  and isomorphisms  $\phi_{ij}: \mathcal{F}_i|_{U_i \cap U_j} \rightarrow \mathcal{F}_j|_{U_i \cap U_j}$  such that  $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$  on  $U_i \cap U_j \cap U_k$  and  $\phi_{ii} = 1_{\mathcal{F}_i}$ . Then there is a unique sheaf  $\mathcal{F}$  on  $X$  with isomorphisms  $\psi_i: \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$  such that  $\psi_j = \phi_{ij} \circ \psi_i$  on  $U_i \cap U_j$ .

*Proof.* The sheaf  $\mathcal{F}$  is quite simple in its definition. For any open subset  $V$  of  $X$ , we define a section of  $\mathcal{F}(V)$  to be a family  $\{s_i\}$  such that  $s_i \in \mathcal{F}_i(V \cap U_i)$  and  $s_i|_{V \cap U_i \cap U_j} = \phi_{ji}(s_j|_{V \cap U_i \cap U_j})$  for all  $i, j$ . It is easy to see that this defines a sheaf, and the existence of the desired isomorphisms is clear. To see that it is unique, let  $\mathcal{G}$  be another such sheaf, and denote the isomorphisms  $\mathcal{G}|_{U_i} \rightarrow \mathcal{F}_i$  by  $\psi'_i$ . Then we have a family of isomorphisms  $\mathcal{G}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$ , which we may glue to an isomorphism  $\mathcal{G} \rightarrow \mathcal{F}$ .  $\square$

**Exercise 2.2.9.** Let  $\mathcal{F}$  be a sheaf on  $X$  whose support  $Y$  is a finite set of closed points  $y$ . Then for any open set  $U$  of  $X$ ,  $\mathcal{F} = \bigoplus_{x \in U \cap Y} \mathcal{F}_x$ .

*Proof.* Let  $U$  be an open subset of  $X$  meeting  $Y$  at only a single point  $y$ . Then for any open neighborhood  $V$  of  $y$  contained in  $U$ , we have  $U = (U \setminus \{y\}) \cup V$ . Since  $\mathcal{F}(U \setminus \{y\}) = 0$ , the value of any section  $s \in \mathcal{F}(U)$  is determined by its restriction to  $V$ , hence by its stalk at  $y$ . The statement now follows from an easy gluing argument.  $\square$

**Exercise 2.2.10.** Let  $(X, \mathcal{O}_X)$  be a ringed space. Let  $Y$  be a subset of  $X$  endowed with the induced topology. Let  $i: Y \hookrightarrow X$  denote the canonical inclusion. Then  $(Y, i^{-1}\mathcal{O}_X)$  is a ringed space.

*Proof.* For any point  $y \in Y$ , we have

$$(i^{-1}\mathcal{O}_X)_y = \varinjlim_{y \in U} ((U' \mapsto \varinjlim_{i(U') \subseteq V} \mathcal{O}_X(V))^\dagger)(U).$$

Since the stalks of the sheafification of a presheaf are the same as the stalks of the presheaf itself, we can simplify this to

$$\varinjlim_{y \in U} \varinjlim_{i(U) \subseteq V} \mathcal{O}_X(V).$$

It is easy to see that this is just the stalk of  $i(y)$  in  $X$ .  $\square$

## 2.3 Schemes

**Exercise 2.3.1.** Let  $X$  be an open subscheme of an affine scheme  $Y$ . The canonical morphism  $X \rightarrow Y$  corresponds by the map  $\rho$  of Proposition 25 to the restriction  $\mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$ .

*Proof.* In the case where  $X$  is a principal open subset, this is just Lemma 3.7. Otherwise, suppose that  $X = \bigcup D(f_i)$  for some family of elements  $f_i \in \mathcal{O}_Y(Y)$ . The inclusions induce a family of maps  $\mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(D(f_i)) = \mathcal{O}_X(X)_{f_i}$  which clearly agree on the intersections. Gluing these gives the restriction.  $\square$

**Exercise 2.3.2.** Let  $U = \text{Spec } B$  be an affine open subscheme of  $X = \text{Spec } A$ . Then the restriction  $A \rightarrow B$  is flat.

*Proof.* Since the restriction corresponds to the open immersion  $U \hookrightarrow X$ , we see that its localization  $A_x \rightarrow B_x$  is an isomorphism, which is flat, for every  $x \in \text{Spec } B$ .  $\square$

**Exercise 2.3.3.**

- (a) Let  $F$  be a subset of an affine scheme  $\text{Spec } A$ . The closure  $\overline{F}$  of  $F$  in  $\text{Spec } A$  is equal to  $V(I)$ , where  $I = \bigcap_{\mathfrak{p} \in F} \mathfrak{p}$ .
- (b) Let  $\varphi: A \rightarrow B$  be a ring homomorphism. Let  $\overline{f}: \text{Spec } B \rightarrow \text{Spec } A$  be the morphism of schemes associated to  $\varphi$ . Then  $\overline{\text{Im } f} = V(\text{Ker } \varphi)$ .

*Proof.* We already have that  $\overline{F}$  must already be of the form  $V(I')$  with  $I' \subset I$ . Since  $V(I')$  must be the minimal such set, we have  $I' = I$ .

For part (b) we assume that  $A$  and  $B$  are reduced, since that doesn't affect their prime spectra. Then the previous part implies that  $\overline{\text{Im } f} = V(\bigcap_{\mathfrak{p} \in \text{Spec } A} \varphi^{-1}(\mathfrak{p})) = V(\text{Ker } \varphi)$ .  $\square$

In the case where  $B = A_{\mathfrak{p}}$  for some  $\mathfrak{p} \in \text{Spec } A$ , this is the set of ideals that are "large enough" to interact with  $\mathfrak{p}$ . In geometric terms, they are the irreducible varieties such that every function that vanishes on  $V(\mathfrak{p})$  vanishes here as well.

**Exercise 2.3.4.** Let  $X$  be a scheme and  $f \in \mathcal{O}_X(X)$ . Then the map  $U \mapsto f|_U \mathcal{O}_X(U)$  for every affine open subset  $U$  defines a sheaf of ideals on  $X$ , denoted  $f\mathcal{O}_X$ . In addition,  $\text{Supp } f\mathcal{O}_X$  is closed.

*Proof.* It is easy to see that  $f\mathcal{O}_X$  inherits the uniqueness condition from  $\mathcal{O}_X$ . On the other hand, suppose that we have an open set  $U$  of  $X$ , an affine open cover  $\{U_i\}$  of  $U$ , and a family of sections  $f|_{U_i} s_i$  that agrees on intersections. Then we may glue these into a section  $s$  of  $\mathcal{O}_X(U)$  that restricts to  $f|_{U_i} s_i$  on every  $U_i$ , which we contest is contained in  $f\mathcal{O}_X(U)$ . Indeed, suppose that  $s$  is not in this ideal. Then for any  $a \in \mathcal{O}_X(U)$ , there is some  $x \in U$  such that  $s_x - (af)_x \neq 0$ , but this contradicts the fact that  $f|_{U_i} s_i$  is in  $f\mathcal{O}_X(U)$ .

To see that  $\text{Supp } f\mathcal{O}_X$  is closed, observe that  $\text{Supp } f\mathcal{O}_X$  is just the set of  $x \in X$  such that  $f_x \neq 0$ . Since the complement is open, the statement holds.  $\square$

**Exercise 2.3.5.** Let  $Y$  be a scheme such that  $\text{Mor}(X, Y) \cong \text{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$  for any affine scheme  $X$ . Then  $Y$  is affine.

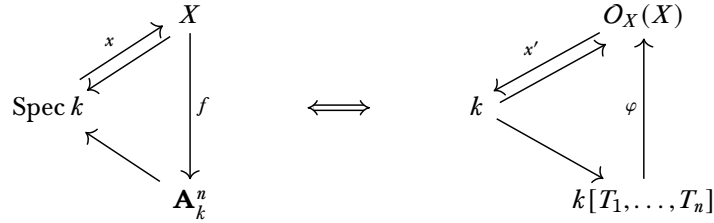
*Proof.* Regrettably, the easiest proof I could come up with was by category slop. Observe that for any affine scheme  $Z$ , we have  $\text{Mor}(Z, Y) \cong \text{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \cong \text{Mor}(Z, \text{Spec}(\mathcal{O}_Y(Y)))$ . This extends to the category of all schemes in the obvious way, so  $Y$  is a representing object of  $\text{Mor}(-, \text{Spec}(\mathcal{O}_Y(Y)))$  and is thus isomorphic to  $\text{Spec}(\mathcal{O}_Y(Y))$ .  $\square$

**Exercise 2.3.6.** Let  $A$  be a ring and consider the  $A$ -scheme  $X = \operatorname{Spec}(A[T_1, \dots, T_n]/I)$ . The  $A$ -points of  $X$  are in bijective correspondence with the  $n$ -tuples  $\alpha = (\alpha_1, \dots, \alpha_n) \in A^n$  such that  $P(\alpha) = 0$  for all  $P \in I$ .

*Proof.* An  $A$ -point of  $X$  is the same as a homomorphism  $A[T_1, \dots, T_n]/I \rightarrow A$  that restricts to the identity on  $A$ , so it is determined only by the values of the  $T_i$ . This homomorphism must send  $T = (T_1, \dots, T_n)$  to a family  $\alpha = (\alpha_1, \dots, \alpha_n)$  satisfying the condition above. The statement is now obvious.  $\square$

**Exercise 2.3.7.** Let  $X$  be a scheme over the field  $k$ . Let  $\varphi: k[T_1, \dots, T_n] \rightarrow \mathcal{O}_X(X)$  be a homomorphism of  $k$ -algebras and  $f: X \rightarrow \mathbf{A}_k^n$  the morphism induced by  $\varphi$ . Then for any rational point  $x \in X(k)$ , via the identification  $\mathbf{A}_k^n = k^n$ , we have  $f(x) = (f_1(x), \dots, f_n(x))$ , where  $f_i = \varphi(T_i)$  and  $f_i(x)$  is the image of  $f_i$  in  $k(x) = k$ .

*Proof.* The statement is obvious if you draw a picture and use the universal property of affine schemes:



$\square$

**Exercise 2.3.8.** Let  $X$  be a scheme over a ring  $A$ . Let  $f_0, \dots, f_n \in \mathcal{O}_X(X)$  be such that the  $f_{i,x}$  generate the unit ideal of  $\mathcal{O}_{X,x}$  for all  $x \in X$ . Then  $X$  is the union of the  $X_{f_i}$ . Moreover, there is a morphism  $f: X \rightarrow \mathbf{P}_A^n$  such that  $f^{-1}(D_+(T_i)) = X_{f_i}$  and  $f|_{X_{f_i}}$  is induced by the homomorphism  $A[T_i^{-1}T_j]_j \rightarrow \mathcal{O}_X(X_{f_i})$  given by  $T_i^{-1}T_j \mapsto f_i^{-1}f_j$ .

*Proof.* Let  $x \in X$  be any point. Since the  $f_{i,x}$  generate the unit ideal, there is  $j$  such that  $f_{j,x}$  is not contained in the maximal ideal at  $x$ , so  $x \in X_{f_j}$ . On the other hand, we have a homomorphism  $A[T_0, \dots, T_n] \rightarrow \mathcal{O}_X(X)$  given by  $T_i \mapsto f_i$ . This gives rise to a homomorphism  $A[T_i^{-1}T_j]_j \rightarrow \mathcal{O}_X(X)_{f_i} \cong \mathcal{O}_X(X_{f_i})$  for any  $i$ , hence a morphism  $X_{f_i} \rightarrow D_+(T_i)$ . We contest that this agrees on intersections. Indeed, observe that for any  $i, j$ , the morphisms  $X_{f_i} \rightarrow D_+(T_i)$  and  $X_{f_j} \rightarrow D_+(T_j)$ , when restricted to  $X_{f_i f_j} = X_{f_i} \cap X_{f_j}$ , both factor through  $D_+(T_i) \cap D_+(T_j) = D_+(T_i T_j)$ ; it is then clear that they agree, so we have the desired morphism  $X \rightarrow \mathbf{P}_A^n$ .  $\square$

Note: I omitted the last part of 2.3.8 because it is the same as 2.3.7.

**Exercise 2.3.9.** Let  $B$  be a graded ring. Let  $f \in B$  be a non-nilpotent, homogeneous element of degree 0. Then  $B_f$  possesses a natural grading, and  $D_+(f) = \operatorname{Proj} B_f$ .

*Proof.* The grading is obvious. Recall that a prime ideal only localizes to a prime in  $B_f$  if it does not contain  $f$ . Then  $D_+(f)$  is just  $\text{Proj } B_f$ .  $\square$

**Exercise 2.3.10.** Let  $A$  be a ring,  $X = \mathbf{P}_A^n$ . Then  $\mathcal{O}_X(X) = A$ , and  $X$  is affine iff  $n = 0$ .

*Proof.* Consider the covering  $\{D_+(T_i)\}$  of  $X$ . Then a global section  $f \in \mathcal{O}_X(X)$  must restrict to a family of sections  $f_i = \mathcal{O}_X(D_+(T_i))$  that agree on intersections. In other words, it is identified with a family of functions  $f_i = P_i(T_i^{-1}T_j)_j$  such that  $P_i(T_i^{-1}T_k)_k = P_j(T_j^{-1}T_k)_k$  for all  $i, j$ . This can only happen if all the  $P_i$  are constant, hence  $\mathcal{O}_X(X) = A$ . That  $X$  is affine only when  $n = 0$  follows immediately.  $\square$

**Exercise 2.3.11.** Let  $B = \oplus_{d \geq 0} B_d$  be a graded ring. Let  $e \geq 1$  be an integer.

- (a) Let us denote the graded ring  $\oplus_{d \geq 0} B_{de}$  by  $C$ . Then  $\text{Proj } B \cong \text{Proj } C$ .
- (b) Let us suppose that for every  $d \geq 1$ ,  $B_{ed}$  is generated by  $B_e^d$ , and that  $B_e$  is of finite type over a ring  $A$ . Then  $\text{Proj } B$  is a projective scheme over  $A$ .

*Proof.* For part (a), observe that any homogeneous prime ideal of  $B$  gives rise to one in  $C$  and vice versa, and this correspondence is bijective. If  $f$  is any non-nilpotent homogeneous element of  $B$ , a homogeneous prime ideal  $\mathfrak{p}$  of  $B$  does not contain  $f$  iff it does not contain  $f^e$ , hence  $\text{Proj } B$  and  $\text{Proj } C$  have the same support. It is easy to see that  $C_{(f^e)}$  is isomorphic to  $B_{(f)}$  by multiplying numerators and denominators by a suitable power of  $f$ , so the statement follows.

Part (b) follows from some basic noggin use.  $\square$

**Exercise 2.3.12.** Let  $B = A[X, Y, Z]$  be a polynomial ring. Let  $B_d$  be the sub- $A$ -module of  $B$  generated by elements of the form  $X^a Y^b Z^c$  with  $a + 2b + 3c = d$ .

- (a) The  $B_d$  induce a grading on  $B$ .
- (b) As a module,  $B_6$  is generated by  $X^6, Y^3, Z^2, X^2 Y^2, X^4 Y, X^3 Z$ , and  $XYZ$ . Moreover, for all  $d$ ,  $B_{6d}$  is generated by  $B_6^d$ , so  $\text{Proj } B$  is a closed subscheme of  $\mathbf{P}_A^6$ .

*Proof.* The first part is obvious, and the generating set for  $B_6$  is checked easily. Suppose that  $B_{6(d-1)} = B_6^{d-1}$ . Let  $a, b, c$  be such that  $X^a Y^b Z^c \in B_{6d}$ . If  $a \geq 6$ ,  $X^a Y^b Z^c \in B_6 B_{6(d-1)}$ , so is an element of  $B_6^d$ . Otherwise,  $2b + 3c \equiv -a \pmod{6}$ , so there are non-negative integers  $b', c'$  such that  $b' \leq b$ ,  $c' \leq c$ , and  $a + 2b' + 3c' = 6$ . It follows that  $X^a Y^b Z^c = (X^a Y^{b'} Z^{c'}) (Y^{b-b'} Z^{c-c'})$  is an element of  $B_6 B_{6(d-1)}$ , and so  $B_{6d}$  is generated by  $B_6^d$ . By Exercise 3.11,  $\text{Proj } B \cong \text{Proj } A[X^6, Y^3, Z^2, X^2 Y^2, X^4 Y, X^3 Z, XYZ]$ , and the right hand side is clearly a subscheme of  $\mathbf{P}_A^6$ .  $\square$

**Exercise 2.3.13.** A scheme is quasi-compact if and only if it is a finite union of affine schemes.

*Proof.* Use noggin.  $\square$

**Exercise 2.3.14.** Let  $X$  be a quasi-compact scheme,  $A = \mathcal{O}_X(X)$ . The morphism  $f: X \rightarrow \text{Spec } A$  is dominant.

*Proof.* Let  $\{U_i\}$  be a finite open covering of  $X$  by affines. It is easy to see that  $f$  is obtained by gluing the morphisms  $U_i \rightarrow \text{Spec } A$ , which are in turn induced by the restriction maps  $A \rightarrow \mathcal{O}_X(U_i)$ . If  $g$  is any non-nilpotent element of  $A$ , there is some  $i$  such that  $g|_{U_i}$  is non-nilpotent, since the cover is finite. If  $x$  is any element of  $U_i$  such that  $g|_{U_i}(x) = g(x)$  is non-zero, we see that  $f(x) \in D(g)$  and so  $f(X)$  is dense in  $\text{Spec } A$ .  $\square$

**Exercise 2.3.15.** A scheme  $X$  is locally noetherian if and only if any affine open subscheme of  $X$  is noetherian.

*Proof.* The "only if" direction is clear. On the other hand, suppose that  $X$  is locally noetherian and let  $U$  be an affine open subscheme. Then  $U$  admits an open covering  $\{U_i\}$  by noetherian affines, which we may assume finite by quasi-compactness.  $\square$

**Exercise 2.3.16.** Let  $f: X \rightarrow Y$  be a morphism of schemes.

- (a) If  $f$  is a closed immersion, it is quasi-compact.
- (b) If  $f$  is an open immersion and  $Y$  is locally noetherian, then  $f$  is quasi-compact.

Let us suppose in what follows that  $f$  is quasi-compact.

- (c) Let  $I = \text{Ker}(f^\#: \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X)$ . The ringed subspace  $Z = V(I)$  of  $Y$  is a scheme.
- (d) Let  $j: Z \rightarrow Y$  be the closed immersion. Then we have a morphism  $g: X \rightarrow Z$  such that  $f = j \circ g$ . Moreover, if  $f$  decomposes into a morphism  $g': X \rightarrow Z'$  followed by a closed immersion  $j': Z' \rightarrow Y$ , then  $Z$  is a closed subscheme of  $Z'$ .
- (e) The image  $f(X)$  of  $f$  is dense in  $Z$ . If  $X$  is reduced, then so is  $Z$ .

*Proof.* Part (a) is clear. For part (b), just recall that any open affine subscheme  $U$  of  $Y$  is noetherian. It suffices to show part (c) for affine  $Y$ . Indeed, if  $\{U_i\}$  is a covering of  $Y$  by affine opens, we get an induced covering of  $Z$  (not necessarily by affines). If each  $U_i \cap Z$  is a scheme, then we can just paste the coverings. Now, assume  $Y$  is affine. The quasi-compactness hypothesis implies that  $X$  is quasi-compact, so it admits a finite open covering  $\{X_i\}$  by affines. Since the canonical map  $\mathcal{O}_X(X) \rightarrow \oplus \mathcal{O}_X(X_i)$  is an injection,  $I = \text{Ker}(f^\#)(X) = \text{Ker}(A \rightarrow \oplus \mathcal{O}_X(X_i))$ .

We contest that  $I$  defines the sheaf  $\text{Ker}(f^\#)$ . Indeed, the flatness of  $A_g$  for any  $g \in A$  gives us a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & A & \longrightarrow & \oplus \mathcal{O}_X(X_i) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & I_g & \longrightarrow & A_g & \longrightarrow & \oplus (\mathcal{O}_X(X_i) \otimes A_g) \end{array}$$

where the rows are exact. Since the  $X_i$  are affine, the bottom right homomorphism corresponds to the restriction  $f^{-1}(D(f)) \rightarrow D(f)$  of  $f$ , and so  $\text{Ker}(f^\#)(D(f)) = I_f$ . It follows that  $Z \cong \text{Spec } A/I$

For part (d), let  $g: X \rightarrow Z$  be the map (not morphism) such that  $g(x)$  is the unique  $z \in Z$  such that  $f(x) = j(z)$ . It is continuous. For any affine open  $U$  of  $Y$ , define  $g^\#(U \cap Z): \mathcal{O}_Z(U \cap Z) \rightarrow (g_*\mathcal{O}_X)(U \cap Z)$  to be the homomorphism such that  $\mathcal{O}_Y(U) \rightarrow (f_*\mathcal{O}_X)(U) = (g_*\mathcal{O}_X)(U)$  factors through  $g^\#(U \cap Z)$  by the quotient  $\mathcal{O}_X(U) \rightarrow \mathcal{O}_Z(U \cap Z) = \mathcal{O}_X(U)/\text{Ker}(f^\#(U))$ . It is easy to see that this defines a morphism of schemes  $(g, g^\#): X \rightarrow Z$ . If  $X \xrightarrow{g'} Z' \xrightarrow{j'} Y$  is any other such factorization, observe that every  $y \in Y$  such that  $\text{Ker}(f^\#)_y \neq \mathcal{O}_{Y,y}$  must be contained in  $Z'$ , since any such  $y$  is an accumulation point of  $f(X)$ . It is easy to see that this implies that  $Z = V(\text{Ker}(g^\#: \mathcal{O}_{Z'} \rightarrow g'_*\mathcal{O}_X))$ .

Part (e) follows from the same accumulation point argument.  $\square$

**Exercise 2.3.17.** Let  $X$  be an affine algebraic variety over a field  $k$ . There is a projective variety  $\bar{X}$  over  $k$  such that  $X$  is isomorphic to a dense open subscheme of  $\bar{X}$ .

*Proof.* Let  $X = \text{Spec } A = \text{Spec } k[T_1, \dots, T_n]/I$ . It is easy to see that embeds as a closed subscheme of  $D_+(T_0) \subseteq \mathbf{P}_k^n$ , so let  $J$  be the kernel of the corresponding ring homomorphism  $k[T_0^{-1}T_1, \dots, T_0^{-1}T_n] \rightarrow A$ . We may extend  $J$  to a homogeneous ideal  $J' = k[T_0, \dots, T_n, T_0^{-1}]J \cap k[T_0, \dots, T_n]$ , so we have a projective variety  $V_+(J')$ . We contest that  $D_+(T_0)$  is dense in  $V_+(J')$ . Indeed, let  $P$  be any homogeneous polynomial in  $k[T_0, \dots, T_n] \setminus \sqrt{J'}$ . Then  $D_+(P) \cap D_+(T_0) \cap V_+(J') = D_+(T_0P) \cap V_+(J')$ , which is nonempty since  $J'$  is not maximal (it does not contain  $T_0$ ).  $\square$

## 2.4 Reduced schemes

**Exercise 2.4.1.** Let  $k$  be a field and  $P \in k[T_1, \dots, T_n]$ . Then  $\text{Spec}(k[T_1, \dots, T_n]/(P))$  is a reduced (resp. irreducible; integral) if and only if  $P$  is squarefree (resp. admits only one irreducible factor; is irreducible).

*Proof.* dawg  $\square$

**Exercise 2.4.2.** Let  $X$  be a scheme and  $x \in X$ . Then the image of the morphism  $\text{Spec } \mathcal{O}_{X,x} \rightarrow X$  is the set of points of  $X$  that specialize to  $x$ .

*Proof.* It suffices to prove this in the case where  $X$  is affine. If  $y$  is any point of  $X$  that specializes to  $x$ , we have  $\mathfrak{j}_y \subseteq \mathfrak{j}_x$  and so  $y$  is in the image of  $\text{Spec } \mathcal{O}_{X,x}$ . The other direction is basically the same.  $\square$

**Exercise 2.4.3.** Let  $\mathcal{O}_K$  be a discrete valuation ring with field of fractions  $K$  and uniformizing parameter  $t$ . Then  $\text{Spec } K[T]$  embeds as an open subscheme of  $\text{Spec } \mathcal{O}_K[T]$ .

*Proof.* Observe that  $(\mathcal{O}_K)_t = K$ , so  $(\mathcal{O}_K[T])_t = K[T]$ . Since  $\text{Spec}((\mathcal{O}_K[T])_t)$  is homeomorphic to  $D(t) \subseteq \text{Spec}(\mathcal{O}_K[T])$ , this proves the proposition.  $\square$

The second part of this exercise is to find the points of  $\text{Spec } K[T]$  which specialize to the point corresponding to the maximal ideal  $(T, t)$  of  $\mathcal{O}_K[T]$ . To do this, notice that  $\text{Spec } K[T]$  corresponds to the ideals  $I$  of  $\mathcal{O}_K[T]$  such that  $I \cap (t) = \{0\}$ , i.e. the principal ideals generated by nonconstant polynomials. Then the only points that specialize to  $(T, t)$  are  $(0)$  and  $(T)$ .

**Exercise 2.4.4.** Let  $X$  be a scheme having only a finite number of irreducible components  $\{X_i\}$ . Then  $X$  is connected if and only if for any pair  $i, j$ , there exist indices  $i_0 = i, i_1, \dots, i_r = j$  such that  $X_{i_l} = X_{i_{l+1}}$  for every  $l < r$ . Furthermore,  $X$  is integral if and only if it is connected and integral at every point.

*Proof.* Since the irreducible components of  $X$  are finite in number, they must all be open; it is clear that having  $X$  connected implies the condition. On the other hand, suppose that  $X$  has the property. If  $U$  is a nonvoid open set such that  $\overline{U} = U$ , we have that  $U$  is the union of irreducible components of  $X$ , so suppose that  $X_i$  is any component (not necessarily contained in  $U$ ). Then we have a sequence  $X = X_0, \dots, X_r \subseteq U$  such that  $X_i \cap X_{i+1} \neq \emptyset$ . However, since  $X_{r-1} \cap U \neq \emptyset$ ,  $X_{r-1}$  is contained in  $U$  and so we may drop the last index. Repeating this, we may assume that  $r = 0$  and so  $X \subseteq U$ .

Now, if  $X$  is integral, it is obvious that it is connected and integral at every point, so assume the latter. At any point  $x \in X$  let  $i: \text{Spec } \mathcal{O}_{X,x} \rightarrow X$  be the canonical morphism. Since  $i^{-1}(i((0))) \subseteq i^{-1}(\overline{i((0))})$ , we have that  $\text{Spec } \mathcal{O}_{X,x} \subseteq i^{-1}(\overline{i((0))})$  and so the image of  $i$  has a closed point. It follows from exercise 2 that each point  $x$  is contained in exactly one irreducible component, so the connectedness hypothesis implies that  $X$  is irreducible. That  $X$  is reduced is obvious, so  $X$  is integral.  $\square$

**Exercise 2.4.5.** Let  $X$  be a scheme. Every irreducible component of  $X$  is contained in a connected component. If  $X$  is locally noetherian, then the connected components are open. If  $X$  is noetherian, the connected components are finite in number.

*Proof.* The first and third statements are clear, and the second one follows from the third.  $\square$

**Exercise 2.4.6.** Let  $A$  be a ring and  $X$  be a scheme.

- (a) The following properties are equivalent:
- (i)  $X$  is connected;
  - (ii)  $\mathcal{O}_X(X)$  has no idempotent elements other than 1;
  - (iii)  $\text{Spec}(\mathcal{O}_X(X))$  is connected.
- (b) Any local scheme is connected.
- (c) Suppose that the connected components of  $X$  are open. Let  $U$  be a connected component of  $X$ . Then there is a unique idempotent element  $e$  of  $\mathcal{O}_X(X)$  such that  $e|_U = 1$  and  $e|_{X \setminus U} = 0$ . This establishes a canonical bijection from the set of connected components of  $X$  onto the set of indecomposable idempotent elements of  $\mathcal{O}_X(X)$ , the inverse map being given by  $e \mapsto V(1 - e)$ .

*Proof.* For part (a), let  $e \neq 1 \in \mathcal{O}_X(X)$  be an idempotent element. At any point  $x \in X$ ,  $e_x$  must also be idempotent, so generates an ideal  $(e_x)$  that squares to itself. If  $e_x \neq 1$ , Nakayama's lemma implies that it is 0. We see that the sets  $U = \{x \in X: e_x = 1\}$  and  $V = \{x \in X: e_x = 0\}$  are disjoint and form an open cover of  $X$ , so (i) implies (ii).

On the other hand, suppose that  $U, V$  are disjoint open sets that cover  $\text{Spec}(\mathcal{O}_X(X))$ . Then there is an element of  $\mathcal{O}_X(X)$  that restricts to 0 on  $U$  and 1 on  $V$ . It is clearly idempotent and not equal to 1, so (ii) implies (iii).

Finally, suppose that  $X$  is disconnected. Then the same argument as in part (ii) implies that there is an idempotent  $e \neq 1 \in \mathcal{O}_X(X)$ , hence  $\text{Spec}(\mathcal{O}_X(X))$  is disconnected and so (a) follows.

Part (b) follows from the same reasoning as part (a)(ii), and part (c) is obvious.  $\square$

**Exercise 2.4.7.** The underlying topological space of a scheme is a Kolmogorov space.

*Proof.* It suffices to show the statement for affine schemes, so let  $X$  be an affine scheme and let  $x$  and  $y$  be distinct points of  $X$ . If  $\mathfrak{j}_x \subset \mathfrak{j}_y$ ,  $X \setminus V(\mathfrak{j}_y)$  is a neighborhood of  $x$  not containing  $y$ . Otherwise, suppose there is no inclusion relation between  $\mathfrak{j}_x$  and  $\mathfrak{j}_y$ . Then there is some  $f \in \mathfrak{j}_x$  such that  $f \notin \mathfrak{j}_y$ , so  $D(f)$  is a neighborhood of  $y$  not containing  $x$ .  $\square$

**Exercise 2.4.8.** A quasicompact scheme has a closed point.

*Proof.* Let  $X$  be a scheme with a finite cover  $\{U_i\}$  by affine opens (assuming no inclusion relation between the  $U_i$ ) and suppose that  $x \in U_1$  is a closed point in  $U_1$  (but not necessarily in  $X$ !). If  $x$  is a closed point in  $X$ , we are done; otherwise, there is another affine open in the cover, say  $U_2$ , such that  $\overline{\{x\}}$  meets  $U_2$  at a point other than  $x$ . It is clear that  $\overline{\{x\}}$  contains a closed point of  $U_2$  which is not contained in  $U_1$ . Continuing this process will give the desired point.  $\square$

**Exercise 2.4.9.** Let  $X$  be a noetherian scheme. Then the set of points such that  $\mathcal{O}_{X,x}$  is reduced (resp. is an integral domain) is open.

*Proof.* First, assume  $X$  is the prime spectrum of a noetherian ring. Denote by  $N$  the nilradical of  $\mathcal{O}_X(X)$ , and let  $f_1, \dots, f_n$  be generators of  $N$ . Then for any  $x \in X$ ,  $\mathcal{O}_{X,x}$  is reduced only if  $\text{Ann}(f_i)$  is not contained in  $\mathfrak{j}_x$  for all  $i$ , i.e. only if  $x \in D(f_1 f_2 \dots f_n)$ . An easy gluing argument gives the statement for general  $X$ .  $\square$

## 2.5 Dimension

**Exercise 2.5.1.** Let  $X$  be a topological space. Let  $\{X_i\}_i$  be a locally finite covering of  $X$  by closed subsets  $X_i$ . Then  $\dim X = \sup_i \dim X_i$ .

*Proof.* Let  $x$  be a point of  $X$  of dimension  $n$ . Choosing a neighborhood  $U$  of  $x$  that only meets finitely many of the  $X_i$ , let  $Z_0 \subseteq Z_1 \subseteq \dots \subseteq Z_n$  be an ascending chain of irreducible subsets. If  $X_1, \dots, X_m$  are the  $X_i$  that meet  $Z_n$ , it is clear that at least one of the  $X_i$  contains  $Z_n$ , hence  $\sup_i \dim X_i \geq n$  and the proposition holds.  $\square$

**Exercise 2.5.2.** Let  $X$  be a scheme and  $Z$  be a closed subset of  $X$ . Then for all  $x \in X$ , we have  $\text{codim}(\overline{\{x\}}, X) = \dim \mathcal{O}_{X,x}$  and  $\text{codim}(Z, X) = \min_{z \in Z} \dim \mathcal{O}_{X,z}$ .

*Proof.* Let  $\overline{\{x\}} \subseteq Z_0 \subseteq \dots \subseteq Z_n$  be an ascending chain of irreducible closed subsets of  $X$ . This induces a chain of specializations  $z_n \rightsquigarrow \dots \rightsquigarrow z_0 \rightsquigarrow x$ , so all of the  $z_i$  lie in the image of  $\text{Spec } \mathcal{O}_{X,x}$  and  $\dim \mathcal{O}_{X,x} \geq \text{codim}(\overline{\{x\}}, X)$ . The other direction is obvious.

The second statement follows from the same argument as above.  $\square$

**Exercise 2.5.3.** We have the following:

- (a) Let  $Z$  be a closed subset of a topological space  $X$ . Then  $\text{codim}(Z, X) = 0$  iff  $Z$  contains an irreducible component of  $X$ . This does not mean that  $\dim Z$  is equal to  $\dim X$ .
- (b) Let  $X = \text{Spec } \mathcal{O}_K[T]$ , where  $\mathcal{O}_K$  is a discrete valuation ring with uniformizing parameter  $t \neq 0$ . Let  $f = tT - 1$ . Then  $(f)$  is a maximal ideal. Let  $x \in X$  be the corresponding point. Then  $X$  is irreducible,  $\dim \mathcal{O}_{X,x} = 1$ , and  $\text{codim}(\{x\}, X) + \dim \{x\} < \dim X$ .

*Proof.* Part (a) is obvious. For part (b), let  $a_0 \in \mathcal{O}_K$  be contained in  $(t)$ . Since  $a_0 = ut^n$  for some unit  $u$ , we have  $1 = (1 + \dots + tT^{n-1})(1 - tT) + u^{-1}a_0$  and so  $(f, a_0)$  is the unit ideal. The case where  $a_0$  is replaced with a general polynomial follows easily, so  $(f)$  is a maximal ideal. That  $X$  is irreducible is a consequence of the fact that  $\mathcal{O}_K[T]$  is entire. To see that  $\mathcal{O}_{X,x}$  is of dimension 1, just notice that it is a discrete valuation ring. Finally, we have that  $\text{codim}(\{x\}, X) + \dim(\{x\}) = 1$ , but  $\dim(\mathcal{O}_K[T]) = 2$ .  $\square$

**Exercise 2.5.4.** Let  $A$  be a ring and  $\mathfrak{p}_1, \dots, \mathfrak{p}_r$  be prime ideals of  $A$ . Let  $I$  be an ideal of  $A$  contained in none of the  $\mathfrak{p}_i$ . Then  $I$  is not contained in  $\mathfrak{p}_1 \cup \dots \cup \mathfrak{p}_r$ .

*Proof.* Suppose  $r = 2$ . Then if  $x \in I \setminus \mathfrak{p}_1$ , we should assume that  $x \in \mathfrak{p}_1$ . Let  $y$  be in  $I \setminus \mathfrak{p}_2$ . If  $x + y \in \mathfrak{p}_1$ ,  $y \notin \mathfrak{p}_1$  and  $y$  is not contained in either ideal. Otherwise,  $x \in \mathfrak{p}_1$ , contradiction.

Now, assume that the statement holds for  $r - 1$  ideals. Letting  $x_i$  be an element of  $I \setminus (\bigcup_{j \neq i} \mathfrak{p}_j)$ , we see that  $x_1 + x_2 \cdots x_r$  is not contained in any of the ideals.  $\square$

**Exercise 2.5.5.** Let  $A$  be a graded ring and let  $\mathfrak{p}_1, \dots, \mathfrak{p}_r$  be homogeneous prime ideals of  $A$ , none of which contain  $A_+$ . Let  $I$  be a homogeneous ideal generated by homogeneous elements of positive degree, contained in none of the  $\mathfrak{p}_i$ . Then there is a homogeneous element of  $I$  not contained in  $\mathfrak{p}_1 \cup \dots \cup \mathfrak{p}_r$ .

*Proof.* Suppose  $r = 2$ . If  $x \in I \setminus \mathfrak{p}_1$  is homogeneous, suppose that  $x \in \mathfrak{p}_2$ . Let  $y$  be in  $I \setminus \mathfrak{p}_2$ . Since both  $x$  and  $y$  are of non-zero degree,  $x^{\deg y} + y^{\deg x}$  is a homogeneous element. It is easy to see that either  $y$  or the sum above is the desired element.

Assume now that the statement holds for  $r - 1$  ideals. Choosing a homogeneous element  $x_i$  of  $I \setminus (\bigcup_{j \neq i} \mathfrak{p}_j)$  for each  $i$ , we see that  $x_1^{\deg x_2 + \dots + \deg x_r} + (x_2 \cdots x_r)^{\deg x_1}$  is the desired element.  $\square$

## 3 Morphisms and base change

### 3.1 The technique of base change

**Exercise 3.1.1.** Let  $S$ ,  $X$ , and  $Y$  be sets, and equip  $X$  (resp.  $Y$ ) with a map  $\varphi: X \rightarrow S$  (resp.  $\psi: Y \rightarrow S$ ). Let  $p: X \times_S Y \rightarrow X$  and  $q: X \times_S Y \rightarrow Y$  be the restrictions of the projections from  $X \times Y$ . Then for any set  $Z$ , equipped with a pair of maps  $f: Z \rightarrow X$  and  $g: Z \rightarrow Y$  such that  $\varphi \circ f = \psi \circ g$ , there is a unique map  $(f, g)_S: Z \rightarrow X \times_S Y$  such that  $p \circ (f, g)_S = f$  and  $q \circ (f, g)_S = g$ .

*Proof.* The existence (but perhaps not the uniqueness) of  $(f, g)_S$  is clear, so suppose that  $h: Z \rightarrow X \times_S Y$  is such that  $p \circ h = f$  and  $q \circ h = g$ . Then for all  $z \in Z$ ,  $h(z) = (p(h(z)), q(h(z))) = (f(z), g(z)) = (f, g)_S(z)$  and so  $h = (f, g)_S$ .  $\square$

**Exercise 3.1.2.** Let  $f: X \rightarrow Y$  be a morphism of schemes. For any scheme  $T$ , let  $f(T): X(T) \rightarrow Y(T)$  denote the map defined by  $f(T)(g) = f \circ g$ . Then  $f(T)$  is bijective for every  $T$  if and only if  $f$  is an isomorphism.

*Proof.* Yoneda  $\square$

**Exercise 3.1.3.** Let  $f: X \rightarrow Y$  be a dominant morphism of irreducible schemes. Then the generic fiber of  $f$  is dense in  $X$ .

*Proof.* Let  $\zeta_X$  and  $\zeta_Y$  be the generic points of  $X$  and  $Y$ , respectively. Then Proposition 1.16 implies that  $\zeta_X$  is in the generic fiber, so it is dense.  $\square$

**Exercise 3.1.4.** Let  $f: X \rightarrow Y$  be a morphism of affine schemes. Then for any affine open subscheme  $V$  of  $Y$ ,  $f^{-1}(V)$  is affine.

*Proof.* Just recall that  $f^{-1}(V) = X \times_Y V$ , so is affine.  $\square$

**Exercise 3.1.5.** Let  $X$  and  $Y$  be algebraic varieties over a field  $k$ . Then

$$\dim(X \times_k Y) = \dim X + \dim Y.$$

*Proof.* It suffices to prove this for affine varieties. Suppose that  $X = \operatorname{Spec} A$  and  $Y = \operatorname{Spec} B$ . By Corollary 2.5.10,  $\dim X = \operatorname{trdeg}_k A$  and  $\dim Y = \operatorname{trdeg}_k B$ . The usual sorites of tensor products give the result.  $\square$

### 3.2 Applications to algebraic varieties

**Exercise 3.2.1.** Let  $Y$  be a noetherian scheme. Then any  $Y$ -scheme  $X$  of finite type is noetherian. Moreover, if  $Y$  is of finite dimension, then so is  $X$ .

*Proof.* The first part is clear, but I will add details because my teachers always yell at me for lack of rigor. Let  $\{Y_i\}$  be a finite open cover of  $Y$  by noetherian affines. If  $f: X \rightarrow Y$  is the structural morphism, we have a finite cover  $\{X_{ij}\}$  of  $X$  by open affines, with  $X_{ij} \subseteq f^{-1}(Y_i)$  for all  $i, j$ . Since each  $\mathcal{O}_X(X_{ij})$  is a finite type algebra over  $\mathcal{O}_Y(Y_i)$ , a noetherian ring,  $X$  is noetherian. Finally, the dimensions of the  $\mathcal{O}_X(X_{ij})$  are all finite if  $Y$  is finite dimensional by Corollary 2.5.17.  $\square$

**Exercise 3.2.2.** Any open immersion into a locally noetherian scheme is of finite type.

*Proof.* Let  $f: X \rightarrow Y$  be an open immersion into a locally noetherian scheme  $Y$ . It is quasi-compact by Exercise 2.3.17(b), so it suffices to show that the restriction homomorphisms are of finite type, for which the affine noetherian case is enough. Let  $Y = \operatorname{Spec} A$  for  $A$  noetherian. Let  $U = Y \setminus V(I)$  be an affine open subset of  $Y$ . Since  $I$  is finitely generated, say by  $f_1, \dots, f_n$ , the sets  $D(f_1), \dots, D(f_n)$  form a finite open cover of  $U$ , and it is now easy to see that  $\mathcal{O}_Y(U)$  is of finite type over  $A$ .  $\square$